

A Formula-Aware Krivine–Maurey Proof of Stability Implies Oscillation Stability

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Source Question

James Hanson asks in *Indiscernible Subspaces and Minimal Wide Types*, arXiv:2004.03062, Question 4.3:

Does (model theoretic) stability imply oscillation stability? That is to say, if T is a stable Banach theory, is every unary formula oscillation stable on models of T ?

This packet gives a claimed positive answer.

Proof Intuition

The Dvoretzky–Milman route naturally produces finite-dimensional Hilbert pieces on which a formula is nearly constant, but those pieces need not be nested. That is why a direct compactness pullback from an ultrapower is too weak: it would falsely prove oscillation stability for arbitrary uniformly continuous Hilbert-sphere colorings.

The proof below uses a different stable-Banach mechanism. Krivine and Maurey, in Raynaud’s type-space formulation, prove that a stable Banach space contains an ℓ_p -type: in a suitable compact type semigroup, one has

$$c_1\tau * \cdots * c_n\tau = \|(c_1, \dots, c_n)\|_p \tau.$$

The key observation is that this argument can be run in an enriched type space which records not only norm-type coordinates, but also all stable formula coordinates relevant to a fixed unary instance $\varphi(x, a)$. The Raynaud/Krivine–Maurey proof uses only compactness, stability of coordinates, dilation, convolution, minimal closed conic classes, and continuity points. Those hypotheses remain true after adding stable formula coordinates.

Therefore the displayed equality is equality of enriched types. In particular, if $\|(c_i)\|_p = 1$, then the coordinate recording the normalized pullback $\varphi(\sigma_c(\bar{x}), a)$ has the same value as the one-vector coordinate. A block extraction converts the type into an actual basic sequence inside the original subspace. The closed span of that block sequence is the required oscillation-stable subspace.

Definitions

Let $M \models T$ be a Banach structure and let $Y \subseteq M$ be an infinite-dimensional closed linear subspace. A bounded uniformly continuous function f on the unit sphere S_Y is oscillation stable on Y if for every $\varepsilon > 0$ there is an infinite-dimensional closed subspace $Z \subseteq Y$ such that

$$\sup\{|f(u) - f(v)| : u, v \in S_Z\} < \varepsilon.$$

A unary formula $\varphi(x, a)$ is oscillation stable if this holds for $f(x) = \varphi(x, a)$ on every model and every infinite-dimensional subspace.

We use Hanson's Banach-language convention that the language contains normalized linear-combination symbols σ_c . Thus, for each scalar tuple $c = (c_1, \dots, c_n)$, the pullback

$$(x_1, \dots, x_n) \mapsto \varphi(\sigma_c(x_1, \dots, x_n), a)$$

is again a formula. Since T is stable, all such pullbacks are stable.

Referee Check and Repair

The first version of this packet had two compressed steps. The formula-aware Raynaud transfer was stated rather than unpacked, and the final passage from type limits to an actual subspace was described as a subsequence diagonalization. The latter wording is not strong enough: a property which holds only for sufficiently far finite blocks does not automatically control vectors involving the first few vectors of the closed span.

The proof below repairs this by using blocks, not a bare subsequence. Each new block is chosen so that it realizes the same enriched type relative to the finite net of the span already constructed. Thus every finite-dimensional initial span is made almost monochromatic before passing to the closed span.

The Enriched Type Semigroup

Fix a separable infinite-dimensional closed subspace $X \subseteq Y$, a countable rational vector subspace $D \subseteq X$ which is dense in X , and a parameter a . Let Δ be a countable family of one-variable formulas over $D \cup \{a\}$ containing:

- all coordinates $x \mapsto \|\lambda x + d\|$, with $\lambda \in \mathbb{Q}$ and $d \in D$;
- the unit-ball coordinate $x \mapsto \varphi(x, a)$, used only on normalized variables;
- every formula obtained from the preceding coordinates by rational scalar dilation, rational finite sums, the normalized combination symbols σ_c , and continuous connectives.

For formula coordinates whose output is evaluated on the unit ball, the closure is taken through the normalized symbols σ_c . The raw convolution coordinates used to define the type semigroup are the norm-type coordinates on sufficiently large Banach balls; the φ -coordinates are read from the associated normalized pullbacks. We also close Δ under the two-variable pullbacks $\theta(u + v)$ which occur when sums of norm-type coordinates are formed. Since T is stable, every formula in this fragment is stable.

For $x \in D$, write

$$\bar{x} = (\theta(x))_{\theta \in \Delta}.$$

Let $\mathcal{T}$ be the pointwise closure of $\{\bar{x} : x \in D\}$. We call its elements Δ -types. The norm coordinate $\nu(\tau) = \tau(\|x\|)$ defines bounded parts $\mathcal{T}_M = \{\tau : \nu(\tau) \leq M\}$, and each $\mathcal{T}_M$ is compact: the norm coordinates put all representatives in a bounded ball of X , while all formula coordinates are uniformly bounded on bounded sorts.

Lemma 1 (enriched convolution). *For $\tau, \rho \in \mathcal{T}$ and $\alpha \in \mathbb{Q}$, choose defining sequences $(x_i), (z_j)$ in D . The formulas*

$$\alpha\tau = \lim_i \overline{\alpha x_i}, \quad \tau * \rho = \lim_i \lim_j \overline{x_i + z_j}$$

define elements of $\mathcal{T}$, independent of the defining sequences. Convolution is commutative and associative, dilation distributes over convolution, and for every rational tuple $c = (c_1, \dots, c_n)$ the map

$$\widehat{T}_c(\tau) = c_1\tau * \dots * c_n\tau$$

is Baire class one and is continuous at a comeager set of points in every closed admissible conic class.

Proof. Fix a coordinate $\theta \in \Delta$. For two defining sequences (x_i) and (x'_i) of the same type and (z_j) , (z'_j) of a second type, the values of $\theta(x_i + z_j)$ and $\theta(x'_i + z'_j)$ are governed by the stable formula $\theta(u + v)$. Grothendieck's double-limit criterion gives equality of the two possible iterated limits, and also permits the order of the two limits to be interchanged. This proves well-definedness and commutativity coordinate by coordinate. Associativity and distributivity follow by applying the same argument to the stable formula $\theta(u + v + w)$ and to rational scalar multiples.

For fixed ρ , if $\tau_m \rightarrow \tau$, choose realizations $x_m \in D$ with $\bar{x}_m$ close to τ_m and $\bar{x}_m * \rho$ close to $\tau_m * \rho$. Then (x_m) defines τ , so $\tau_m * \rho \rightarrow \tau * \rho$. Thus convolution is separately continuous. On the compact annuli $\mathcal{T}_{M+1} \setminus \text{int } \mathcal{T}_M$, separately continuous real coordinates are Baire class one; since the fragment is countable, convolution and all countably many maps $\widehat{T}_c$ have a common comeager set of continuity points in each closed conic class. $\square$

The Formula-Aware Raynaud Type

A type is *symmetric* if $(-1)\tau = \tau$. A nonzero closed set $\mathcal{C}$ of symmetric types is a *conic class* if it is closed under rational dilations and convolution. It is *admissible* if it contains a type with positive norm coordinate.

Proposition 1 (formula-aware Raynaud type). *There are $1 \leq p < \infty$ and a symmetric admissible Δ -type τ , realized by blocks from X , such that after normalizing $\nu(\tau) = 1$,*

$$\widehat{T}_c(\tau) = \|c\|_p \tau \tag{2}$$

for every rational finite scalar tuple c .

Proof. We give the exact Raynaud–Krivine–Maurey argument, noting at each point why the added formula coordinates do not change it.

First, symmetric admissible types exist. Take a bounded separated sequence (u_i) in D . Any cluster type q gives $q*(-q)$, the type of differences $u_i - u_j$, and this type is symmetric and has positive norm coordinate. Intersections of chains of closed admissible conic classes remain admissible: scale an admissible element into a fixed compact norm annulus and use the finite intersection property. Therefore Zorn's lemma gives a minimal closed admissible conic class $\mathcal{C}$. By Lemma 1, the countable family of maps $\widehat{T}_c$, $c \in \mathbb{Q}^{<\mathbb{N}}$, has a common point of continuity $q \in \mathcal{C}$. Replacing q by a rational dilation if necessary, assume $\tau = q * (-q)$ is admissible.

Let $(x_i) \subseteq D$ define q , and let $y_i = x_{2i-1} - x_{2i}$ define τ . Passing to a subsequence, form the usual spreading model E of (y_i) , with spreading basis (e_i) . A rational vector $\sum_{i=1}^n b_i e_i$ is said to realize $b_1\tau * \dots * b_n\tau$. This realization convention is compatible with convolution by construction.

The important comparison estimate is the following. If rational vectors $u, v \in \text{span}_{\mathbb{Q}}\{e_i\}$ realize enriched types ρ_u, ρ_v , then for every coordinate $\theta \in \Delta$,

$$|\rho_u(\theta) - \rho_v(\theta)| \leq \omega_\theta(\|u - v\|_E), \quad (3)$$

where ω_θ is a modulus of uniform continuity for θ on a bounded sort containing the relevant representatives. Indeed, write u and v with the same number of coordinates, realize them by far-out blocks of the defining sequence (y_i) , apply the uniform continuity of θ in the original Banach norm, and pass to the iterated limits which define both the spreading norm and the enriched types. This is the point at which formula coordinates enter the proof. When θ is a unit-ball formula coordinate, it is first composed with the appropriate normalized combination symbol σ_c ; hence all evaluations remain in the correct sort, and the same uniform-continuity estimate applies.

For a rational tuple $c = (c_1, \dots, c_m)$, define the block operator

$$A_c e_k = \sum_{j=1}^m c_j e_{m(k-1)+j}.$$

Then $A_c u$ realizes $\widehat{T}_c(\rho_u)$. The operators A_c commute up to the tensor product rule

$$A_c A_d = A_{c \otimes d}, \quad \widehat{T}_c \widehat{T}_d = \widehat{T}_{c \otimes d}.$$

The classical Krivine–Maurey approximate-eigenvector lemma for the spreading basis says that for every finite commuting family $c^{(1)}, \dots, c^{(N)}$ there are numbers $\beta_1, \dots, \beta_N$ and normalized rational vectors $h_m \in E$ such that

$$\|A_{c^{(\ell)}} h_m - \beta_\ell h_m\|_E \longrightarrow 0 \quad (1 \leq \ell \leq N), \quad (4)$$

with $\|c^{(\ell)}\|_\infty \leq \beta_\ell \leq \|c^{(\ell)}\|_1$. This is the finite joint approximate point-spectrum step in the original Krivine–Maurey proof; it uses only the spreading basis and the Banach-space norm, not the list of type coordinates.

Let ρ_m be the enriched type realized by h_m . The set of all limits of such approximating sequences is a closed admissible conic subclass of $\mathcal{C}$: dilation is immediate, convolution is obtained by putting two realizing vectors on disjoint later supports, and closure follows by a diagonal choice. Minimality of $\mathcal{C}$ therefore implies that the common continuity point q , and hence $\tau = q * (-q)$, is a limit of approximating types satisfying (4).

Now evaluate at τ . Since q was chosen as a common continuity point for all $\widehat{T}_c$, and since (3) turns the norm convergence (4) into coordinatewise convergence for every $\theta \in \Delta$, we obtain

$$\widehat{T}_{c^{(\ell)}}(\tau) = \beta_\ell \tau \quad (1 \leq \ell \leq N) \quad (5)$$

as equality in the enriched type space. Thus every coordinate, including $\varphi(x, a)$, is carried along by the same equality.

Finally define $N(c)$ by $\widehat{T}_c(\tau) = N(c)\tau$. The norm coordinate shows that N is a normalized, permutation-invariant, 1-unconditional norm on rational c_{00} , satisfying

$$N(c \otimes d) = N(c)N(d), \quad \|c\|_\infty \leq N(c) \leq \|c\|_1.$$

The standard Krivine–Maurey multiplicative-norm calculation gives $N(c) = \|c\|_p$ for a single $1 \leq p < \infty$. For completeness, recall the calculation: $N(1, \dots, 1) = n^{1/p}$ follows from tensor multiplicativity and monotonicity by comparing n^k with powers of 2; applying the same argument to tensor powers of an arbitrary positive rational vector and using multinomial counting gives $N(c)^p = \sum_i |c_i|^p$; unconditionality gives signs. This proves (2). $\square$

From The Type To A Real Subspace

Lemma 2 (block extraction). *Let τ be as in Proposition 1, normalized so that $\nu(\tau) = 1$, and put $r = \tau(\varphi(x, a))$. For every $\varepsilon > 0$ there is a normalized basic block sequence $(z_k) \subseteq X$ such that for*

$$Z = \overline{\text{span}}\{z_k : k \in \mathbb{N}\}$$

one has

$$|\varphi(u, a) - r| < \varepsilon \quad (u \in S_Z).$$

Proof. Let $(y_i) \subseteq X$ be a defining block sequence for τ . Equality (2) says that if $c \in \mathbb{Q}^n$ and $\|c\|_p = 1$, then the enriched type of the corresponding normalized block combination, in the iterated block limit, is again τ . In particular, both its norm coordinate and the $\varphi(\sigma_c(\bar{x}), a)$ -coordinate are the same as those of τ .

We construct $z_1, z_2, \dots$ recursively as finite rational blocks of (y_i) , with successive disjoint supports. Fix an increasing exhaustion $\Delta_1 \subseteq \Delta_2 \subseteq \dots$ of the fragment by finite sets of coordinates and choose $\delta_m \downarrow 0$. The induction invariant is stronger than small oscillation: for every rational unit vector $b = (b_1, \dots, b_m)$ in a prescribed finite δ_m -net of the ℓ_p^m -sphere, the normalized block vector $\sigma_b(z_1, \dots, z_m)$ has Δ_m -type within δ_m of τ .

Assume $z_1, \dots, z_{m-1}$ have been chosen. Take a finite rational net for the unit sphere of ℓ_p^m . For a net vector $(b_1, \dots, b_m)$, put $s = \left(\sum_{j < m} |b_j|^p\right)^{1/p}$. By the induction hypothesis, when $s > 0$ the normalized previous part $s^{-1} \sum_{j < m} b_j z_j$ has enriched type close to τ on a finite coordinate set. Equality (2) gives

$$s\tau * b_m\tau = (s^p + |b_m|^p)^{1/p}\tau = \tau$$

for the corresponding model combination. Hence the requirements that $\sigma_b(z_1, \dots, z_m)$ be Δ_m -close to τ , for all net vectors b , form a finite set of open conditions in the enriched type topology. Since τ is realized by arbitrarily far normalized blocks of (y_i) , a far-out rational block can be chosen as z_m satisfying all these conditions. The case $s = 0$ is just the requirement that z_m itself realize τ on Δ_m .

This recursive construction has the following consequence. For each m , every unit vector in the finite-dimensional space $\text{span}(z_1, \dots, z_m)$ has φ -value within $\varepsilon/2$ of r : approximate it by a point of the rational net used at stage m , then use the common modulus of uniform continuity of φ and the fact that the net point has Δ_m -type close to τ . The norm coordinates in the same finite sets make (z_k) a basic sequence, because the finite-dimensional norms are uniformly close to the corresponding ℓ_p -sum norms on the chosen nets.

Now let $u \in S_Z$. Pick m and $u_m \in S_{\text{span}(z_1, \dots, z_m)}$ with $\|u - u_m\|$ so small that uniform continuity changes φ by less than $\varepsilon/2$. The previous paragraph gives $|\varphi(u_m, a) - r| < \varepsilon/2$, hence $|\varphi(u, a) - r| < \varepsilon$. $\square$

Main Theorem

Theorem 1. *Let T be a stable Banach theory. Then every unary formula of T is oscillation stable on models of T .*

Proof. Let $M \models T$, let $\varphi(x, a)$ be a unary formula instance, let $Y \subseteq M$ be an infinite-dimensional closed subspace, and fix $\varepsilon > 0$.

Choose a separable infinite-dimensional closed subspace $X \subseteq Y$, and build the enriched fragment over X and a as above. By Proposition 1 and Lemma 2, there is an infinite-dimensional closed subspace $Z \subseteq X \subseteq Y$ and a real number r such that

$$|\varphi(u, a) - r| < \varepsilon/2 \quad (u \in S_Z).$$

Therefore, for all $u, v \in S_Z$,

$$|\varphi(u, a) - \varphi(v, a)| < \varepsilon$$

so $\varphi(x, a)$ is oscillation stable on Y . Since M, Y, a , and ε were arbitrary, the theorem follows. $\square$

Why The Naive Symmetry Route Is Not Used

The proof does not rely on the false principle that a symmetric uniformly continuous coloring on the Hilbert sphere becomes constant on vectors with small ℓ_∞ -coordinates. That principle fails: if u_n is the flat vector of support n in ℓ_2 , and A is the union of the sign/permutation orbits of u_{2^k} , then $x \mapsto d(x, A)$ is Lipschitz and symmetric but separates the support sizes 2^k and $3 \cdot 2^k$ by a fixed amount.

The present argument uses more than symmetry. It uses stability of every normalized linear-combination pullback, which is precisely what makes the formula coordinate part of the Krivine–Maurey type equality.

Verification Notes

The proof was checked against three prior failed routes:

- The compactness/Dvoretzky–Milman Hilbert-subspace route fails because finite good subspaces need not be nested.
- The minimal-wide-type route isolates the same obstruction as preservation of tail finite satisfiability through the Hilbert Morley spine.
- The bare symmetric-flattening lemma is false, as explained above.

The formula-aware Krivine–Maurey route avoids all three. It builds the sequence inside the original subspace Y , and the formula value is a coordinate of the same stable type equality that produces the ℓ_p spreading behavior.

Human review should focus on Proposition 1 and Lemma 2. Proposition 1 is where the exact Raynaud/Krivine–Maurey conic-class proof is transferred from norm types to the enriched stable type space. Lemma 2 is the repair of the old subsequence diagonalization: it uses blocks so that finite initial spans, and hence the closed span, are controlled.

Novelty Check

Local run indexes were searched for arXiv:2004.03062, Hanson, Question 4.3, minimal wide types, oscillation stability, Krivine–Maurey, and stable Banach theory. Existing durable work contained only a Hilbert-operator partial result, a WAP conditional reduction, and failed attempt notes.

Fresh web searches on 2026-06-16 for exact and close phrases including

- “Does (model theoretic) stability imply oscillation stability”

- “Hanson oscillation stability stable Banach theory”
- “stability implies oscillation stability Banach unary formula”
- “Question 4.3 Indiscernible Subspaces oscillation stability”

found Hanson’s source paper, the absorbing paper, and the Shelah–Usvyatsov minimal-types paper. They did not reveal a later explicit answer to Question 4.3. The proof here appears to be a new synthesis of Hanson’s question with the Krivine–Maurey/Raynaud stable type machinery.

References

References

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